Powers of i and the Complex Plane

Reading powers of the imaginary unit and points on the complex plane

Two models to work from

- **The cycle.** Each new power multiplies the one before it by i, and after four steps you are back where you started. Because of that, a power of i depends only on the remainder when the exponent is divided by 4.
- **The plane.** A complex number $a + bi$ is the point (a, b): a along the horizontal *real* axis, b up the vertical *imaginary* axis. Multiplying by i moves a point a quarter turn counterclockwise around the origin, and its distance from the origin never changes.
- Show your reduction step (for example $i^{27} = i^{24} \cdot i^3$), not just the final value.

1. Complete the cycle table. The first row is filled in for you.

Power	i^1	i^2	i^3
Value	i	________	________

Power	i^4
Value	________

Show the multiplication that produces each entry.

2. Simplify: i^{27}

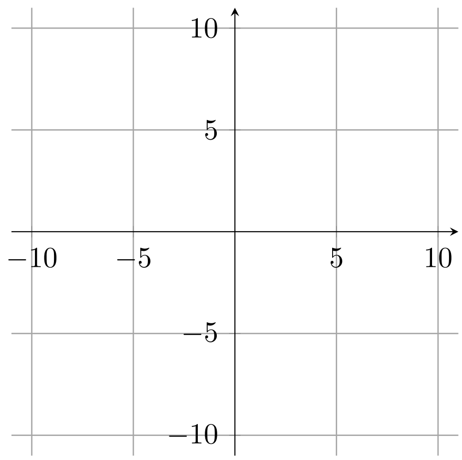
Answer: ________________

3. Let $z = 2 + 5i$.

(a) Write $-z$ and $2z$ in standard form.

$-z =$ __________ $2z =$ __________

(b) Plot and label z, $-z$ and $2z$ on the plane below.



4. Simplify: $i^{40} + i^{22}$

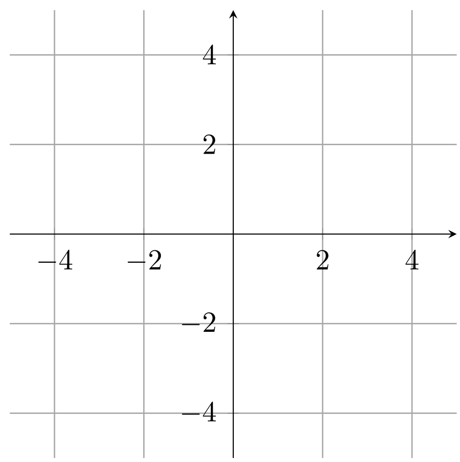
Answer: ____________________

5. Let $z = 3 + i$.

(a) Compute iz and write it in standard form.

$iz =$ _________

(b) Plot z and iz on the plane below, then describe in one sentence what multiplying by i did to the point.



6. Simplify: $3i^{14} - 5i^{7}$

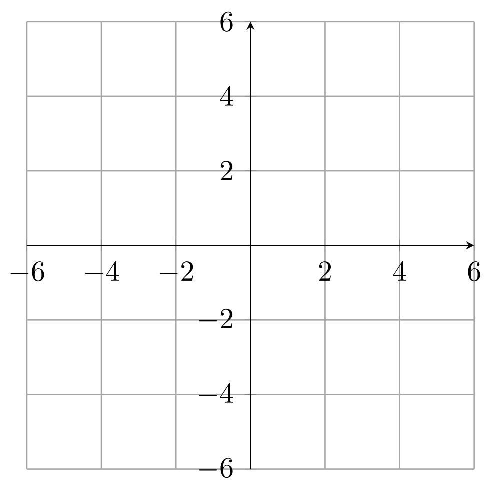
Answer: __________________

7. A point w of the complex plane sits 4 units to the left of the origin and 3 units above it.

(a) Write w in standard form, and give its distance from the origin.

$w =$ __________ distance $=$ __________

(b) Plot w on the plane below and draw the segment joining it to the origin.



8. A point starts at $z = 6 - 2i$ and is turned a quarter turn counterclockwise about the origin twice in a row, which is the same as multiplying by i^2.

new position $=$ __________

Explain in one or two sentences why two quarter turns must negate both the real part and the imaginary part.

9. Simplify: $(2i)^5 + i^{50}$

Answer: ________________

10. A robot arm reaches the point $z = 5 + 2i$. The controller then turns the arm a quarter turn counterclockwise about the origin three times in a row.

(a) Write the arm's final position in standard form.

final position = _________

(b) Plot the start and the finish on the plane below, then justify why three quarter turns give the same result as multiplying by i^3.